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NPTEL (<https://swayam.gov.in/explorer?ncCode=NPTEL>) » Introduction to Machine Learning (course)

Course outline

About NPTEL ()

How does an NPTEL online course work? ()

Week 0 ()

Week 1 ()

Week 2 ()

Week 3 ()

Week 4 ()

- ☐ Separating Hyperplane Approaches - Perceptron Learning (unit? unit=51&lesson=52)
- ☐ Support Vector Machines I - Formulation (unit?

Week 4: Assignment 4 (Non Graded)

Assignment not submitted

Note : This assignment is only for practice purpose and it will not be counted towards the Final score.

1) Which of the following are convex functions?

1 point

☐
 $f(x) = (\sum_{i=1}^n x_i^p)^{1/p}$ where $x \in R^n$ and $p \geq 0$
☐
 $f(x) = \log(\sum_{i=1}^n \exp x_i)$ where $x_i \in R^n$
☐
 $f(x) = \sum_{i=1}^n \sin(x_i)$, where $x \in R^n$
☐
 $f(x) = \sum_{i=1}^n x_i \log x_i$ where $x \in R^n$

2) We discussed two approaches to classification, one that learns the discriminant functions, and the other that is based on modeling hyperplanes. Which of these approaches is more suitable for multi-class problems and why? **1 point**

☐

Discriminant functions; because they allow for a probabilistic interpretation of the predictions.

☐

Hyperplane methods; because they allow for a probabilistic interpretation of the predictions.

☐

Discriminant functions; because an appropriate set of functions will allow us to efficiently disambiguate class predictions.

☐

Hyperplane methods; because we can use basis expansion to transform the input to a space where class-boundaries are linear.

3) Which of the following is/are true regarding an SVM?

1 point

unit=51&lesson=53)

○ Support Vector Machines II - Interpretation and Analysis (unit?

unit=51&lesson=54)

○ SVMs for Linearly Non Separable Data (unit? unit=51&lesson=55)

○ SVM Kernels (unit? unit=51&lesson=56)

○ Hinge Loss formulation of SVM Objective (unit? unit=51&lesson=57)

○ **Practice: Week 4: Assignment 4 (Non Graded) (assessment? name=267)**

● Quiz: Week 4: Assignment 4 (assessment? name=294)

○ Week 4 Feedback Form : Introduction To Machine Learning (unit? unit=51&lesson=285)

● Week 4: Solution (unit? unit=51&lesson=214)

Week 5 ()

Week 6 ()

- ☐ For two dimensional data points, the separating hyperplane learnt by a linear SVM will be a straight line.
- ☐ In theory, a Gaussian kernel SVM can model any complex separating hyperplane.
- ☐ For every kernel function used in a SVM, one can obtain a equivalent closed form basis expansion.
- ☐ Overfitting in an SVM is a function of number of support vectors.

Check Answers and Submit

Your score is: 0/3



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